

Pranav Bose

Embedded Systems & IoT Firmware Engineer • 1 Years Experience • MERIQ Index: 68% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: I2C/SPI/UART, ARM Cortex, MQTT, C/C++, PCB Debugging, FreeRTOS, Python, Embedded Linux

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core I2C/SPI/UART service, reducing latency by 29%.
- Led a squad of 8 engineers delivering scalable microservices processing 12M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 55%.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [I2C/SPI/UART, ARM Cortex, MQTT, C/C++]

Engineered end-to-end production solution leveraging I2C/SPI/UART and ARM Cortex with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [MQTT, C/C++, PCB Debugging, FreeRTOS]

Architected high-resilience system integrating MQTT with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Software Engineering

University of Washington — 2021 - 2025 (Score: 3.45 / 4.0)